of the channel number between the two limiting channels, n_1 and n_2. The net peak area count, N_{peak}, is then calculated from

$$N_{\text{peak}} = \left[\sum_{i=n_1}^{n_2} c(i)\right] - 0.5 \times (n_2 - n_1) \times [c(n_1) + c(n_2)] \tag{9.30}$$

where $c(i)$ is the count in channel i. The uncertainty in N_{peak}, $s_{N_{peak}}$, can be estimated from

$$s_{N_{peak}} = \sqrt{\left[\sum_{i=n_1}^{n_2} c(i)\right] + 0.5 \times (n_2 - n_1) \times [c(n_1) + c(n_2)]} \tag{9.31}$$

9.9. SPECIAL COUNTING SYSTEMS

For *low intensity measurements*, in which a level of radioactivity comparable to the normal background radiation is to be measured, special electronic circuits incorporating two detectors are used. The detectors are coupled so that a signal registers only when both detectors are activated at the same time (*coincidence circuit*) or, alternatively, when only one but not the other is activated at the same time (*anticoincidence circuit*). With such arrangements the normal β, γ background of a detector may be decreased by more than a factor of 100. In the most advanced coincidence techniques both detectors are energy sensitive as well, providing information on the type of radiation being measured. For example, β-γ coincidence measurements are used for absolute determination of radioactivity for samples in which β-decay is immediately followed by γ-emission.

Detectors which are *direction sensitive* have been developed primarily for use in medical diagnosis with radioactive isotopes. The simplest version involves a scintillation detector surrounded by a lead shield with a small hole (collimator) through which radiation reaches the detector. More complex systems with many detector/collimator pairs are in common use (γ-cameras); cf. section 18.5. Scanning instruments (Fig. 18.13) have been developed which permit the measurement of radioactivity as a function of several coordinates as illustrated in Figure 18.14. Such instruments make it easy to detect the accumulation of a radioactive tracer in a particular organ of the body.

Whole-body counters were originally developed for investigation of poisoning by radioactive substances such as radium. They are now used diagnostically and consist of a large scintillation, or semiconductor, detector with the whole system, including the patient, placed in a heavily shielded room. The sensitivity is sufficient to measure natural radioactivity in the human body from such nuclides as ^{40}K.

In nuclear installations like uranium mines, nuclear reactors, reprocessing plants, etc., it is necessary to continuously monitor gas and liquid effluents. Figure 9.23 shows an

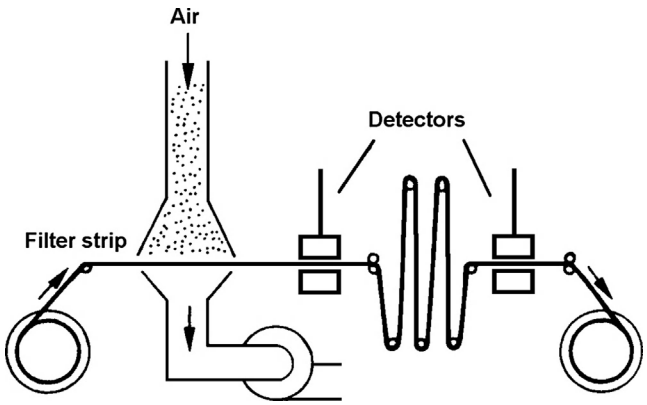


Figure 9.23 Monitor for radioactive aerosols with prompt and delayed measurements. *(From H. Kiefer and R. Maushardt.)*

arrangement for monitoring of *radioactive aerosols* (e.g. Tc, Ru, actinides in air). Two detectors are used, so that some activity (e.g. mother or daughter activity) is allowed to decay between the two detectors. The delay time is adjusted by varying the length of the paper strip between the detector and the rate of movement of the paper strip. The detector may be energy sensitive or simple GM, proportional or scintillation devices. In monitoring of water the detectors (e.g. GM tubes) may dip into the streaming water, or the water may flow around the detector as in Figure 9.9(a).

Radioactive gas flows can be measured by using capillaries or packed columns of 2—20 ml volume containing scintillating material, either for monitoring purposes or, commonly, combined with a gas chromatograph for obtaining "radio-chromatograms" of ^{3}H or ^{14}C labeled organic substances. These detectors, which are commercially available, are viewed by PMTs connected to PC based analysis and display systems.

Radioactive liquid flows can be monitored in several ways depending on the decay mode, energy and liquid film thickness. Liquid flow GM-counters (Fig. 9.9.a), or other types of flow cells combined with solid state detectors, are used at high β- and γ-energies. Glass scintillators embedded in tubings (alternatively tubings containing scintillators) are used for low-energy β-emitters (e.g. ^{3}H, ^{14}C, ^{35}S). The flow cells have typically volumes from 0.01 to 5 ml. By using peak analysis (i.e. analyzing each scintillation intensity versus time, so-called "time-resolved technique") it is possible to measure α-emitters in high β-γ fluxes and even to obtain crude α-spectra, as well as to discriminate different β-emitters and to reduce background. A weakness of this technique is the memory effect in the detectors caused by radionuclide sorption. Therefore, the commercial systems have easily exchangeable flow cells. This demand is circumvented in the LISOL-system, in which the liquid flow is premixed with acid and scintillator solution before counting; the use of that system is described in section 18.4.3.

9.10. ABSOLUTE DISINTEGRATION RATES

The determination of absolute disintegration rates is of great importance in all areas of nuclear chemistry, tracer work, age calculation, etc. Numerous methods have been employed, many using techniques described above, as track counting, liquid scintillation measurements, 4π proportional counters, etc. If the nuclei decay through β-γ emission, the absolute rate may be obtained by two detectors placed close to each side of a thin sample, one detector β-sensitive and the other γ-sensitive.

When only a single detector in a conventional counting set-up is available (e.g. detector arrangement in Fig. 9.4), absolute counting rates can be obtained for unknown samples by comparison with known standards.

When standards are not available it is possible to obtain an approximate estimation of the absolute disintegration rate from a knowledge of the various factors that influence the counting efficiency. The detection efficiency ψ is defined as a ratio between the count rate and the absolute disintegration rate Eqn (5.64). This detection efficiency, which was discussed briefly in section 7.2, is the product of all the factors which influence the measured count rate and may be expressed as

$$\psi = \psi_{\text{det}}\psi_{\text{res}}\psi_{\text{geom}}\psi_{\text{back}}\psi_{\text{self}}\psi_{\text{abs}} \tag{9.32}$$

where

ψ_{det} = counting efficiency of detector,
ψ_{res} = resolving time correction (see section 9.3),
ψ_{geom} = geometry factor (see section 7.2),
ψ_{back} = backscattering factor,
ψ_{self} = self-absorption factor ($\psi_{\text{sample}} = \psi_{\text{back}}\psi_{\text{self}}$), and
ψ_{abs} = absorption factor (see section 7.2).

The efficiency of the detector is a measure of the number of counts registered compared to the number of particles that enter the sensitive volume of the detector. This efficiency is approximately 100% for α- and high-energy β-particles in most detectors, but often substantially lower for γ-rays. In as much as it is quite difficult to apply simple geometric considerations to the solid angle subtended by a detector for a source which is not concentrated at a point the factors ψ_{det} and ψ_{geom} may be determined experimentally by using a very thin standard source of approximately the same area as the unknown. The factor ψ_{geom} can be calculated for circular samples and detector windows, see section 7.2

It was noted in section 7.4 that β-rays undergo large angle deflections. As a result, β-particles from the sample which may start in a direction away from detector can be deflected by several scattering events back into the detector. Such backscattering is dependent upon the atomic number of the material upon which the sample is supported (cf. Fig. 7.14). ψ_{back} increases with backing material thickness up to a saturation thickness beyond which it is constant. Counting is usually done with either an